

Revision — mensuration of solids

Every formula of the block, forwards and backwards, with the units that decide the marks.

LESSON 151–154

4 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 15 CONSOLIDATION

LESSON CLOCK

30'

30'

35'

35'

25'

5

The formulae of the block	30 min
Forward questions	30 min
Reverse questions	35 min
Context and units	35 min
The mistakes that cost marks	25 min
Homework	5 min

LEARNING OBJECTIVES

- State and use every formula of the block from memory.
- Answer forward questions — volume and surface area from the edges.
- Answer reverse questions — an edge or a face from the volume or the surface area.
- Name the five mistakes that lose the most marks and avoid each one.

TEXTBOOK

National	pp. 439–448
Cambridge	Stage 7, pp. 156–172
Workbook	Review 15

01

Explanation

The formulae of the block

WANTED	FOR A CUBOID	FOR A CUBE
volume	$V = abc$	$V = a^3$
surface area	$S = 2(ab + bc + ac)$	$S = 6a^2$
one face	$ab, bc \text{ or } ac$	$\frac{S}{6}$
a missing edge	$\frac{V}{ab}$	the cube root of V
the edge from S	solve $2(ab + bc + ac) = S$	the square root of $\frac{S}{6}$

TWO CONVERSIONS CARRY MOST OF THE MARKS

1 litre = 1000 cm³, and 1 m³ = 1 000 000 cm³ = 1000 litres. Everything else in the block is a multiplication or its reverse.

Forward questions

SOLID	VOLUME	THE THREE FACES	SURFACE AREA
$5 \times 4 \times 3 \text{ cm}$	60 cm^3	20, 12, 15	94 cm^2
$8 \times 5 \times 2 \text{ cm}$	80 cm^3	40, 10, 16	132 cm^2
$10 \times 6 \times 4 \text{ cm}$	240 cm^3	60, 24, 40	248 cm^2
cube, edge 4 cm	64 cm^3	16 each	96 cm^2
cube, edge 7 cm	343 cm^3	49 each	294 cm^2

THREE NUMBERS MULTIPLIED, THREE PRODUCTS DOUBLED

Every forward question in the block is one of those two sentences. Write the three faces on their own line and the surface area cannot go wrong.

Reverse questions

GIVEN	WANTED	WORKING	ANSWER
$V = 120$, base 5×4	the height	$120 \div 20$	6 cm
$V = 240$, base 10×6	the height	$240 \div 60$	4 cm
cube, $V = 512$	the edge	$a^3 = 512$	8 cm
cube, $S = 150$	the edge	$150 \div 6 = 25$	5 cm
$a = 5$, $b = 4$, $S = 94$	the third edge	$40 + 18c = 94$	3 cm

THE DIRECTION DECIDES THE OPERATION

Forward is multiplication; reverse is division, a root, or a linear equation. Deciding which of the two a question is before touching the numbers saves more marks than any amount of arithmetic care.

Context and units

PROBLEM	STEPS	ANSWER
a $60 \times 40 \times 50$ cm tank, in litres	$120\,000 \div 1000$	120 litres
...filled at 8 litres a minute	$120 \div 8$	15 min
water 6 cm deep on a 40×25 cm base	$1000 \cdot 6$	6 litres
a stone raising that level by 1.5 cm	$1000 \cdot 1.5$	1500 cm ³
a $4 \times 3 \times 0.1$ m slab at 2.4 t per m ³	$1.2 \cdot 2.4$	2.88 t

CONVERT FIRST, CALCULATE SECOND

Every context question in this block is a formula from the first section with a unit conversion in front of it. Doing the conversion first turns a hard question into an easy one.

The mistakes that cost marks

MISTAKE	LOOKS LIKE	CORRECT
a metre left as 1	$1 \cdot 80 \cdot 50 = 4000\text{ cm}^3$	$100 \cdot 80 \cdot 50 = 400\,000\text{ cm}^3$
a cube edge by dividing by three	$27 \div 3 = 9$	$a^3 = 27, a = 3$
the root taken before the six	$\sqrt{96}$	$\sqrt{(96 \div 6)} = 4$
litres divided by an area	$120 \div 4000$	$120\,000 \div 4000 = 30\text{ cm}$
a volume written in cm ²	60 cm ²	60 cm ³

FOUR OF THE FIVE ARE UNITS, NOT ARITHMETIC

Which is the good news: reading the units off the working, line by line, removes almost every lost mark in this block.

02 Worked examples

EXAMPLE 1 Find the volume and the surface area of a cuboid 10 cm by 6 cm by 4 cm.

① $V = 10 \cdot 6 \cdot 4 = 240 \text{ cm}^3$.

② Faces: 60, 24, 40.

③ $S = 2 \cdot 124 = 248 \text{ cm}^2$.

Volume cubed, area squared ✓

ANSWER

240 cm³ and 248 cm²

EXAMPLE 2 A cube has surface area 150 cm². Find its edge and its volume.

① $150 \div 6 = 25 \text{ cm}^2$ for one face.

② The edge is 5 cm.

③ $V = 5^3 = 125 \text{ cm}^3$.

Check: $6 \cdot 25 = 150$ ✓

ANSWER

5 cm and 125 cm³

EXAMPLE 3 A tank with base 80 cm by 50 cm holds 120 litres when full. Find its depth.

① 120 litres is $120\,000\text{ cm}^3$.

② Base area $80 \cdot 50 = 4000\text{ cm}^2$.

③ $120\,000 \div 4000 = 30\text{ cm}$.
Convert first, divide second.

ANSWER

30 cm

03 Interactive model

Give the class the five-row mistake table before the revision problems, not after; most of them recognise at least one of their own habits in it.

Revision of the solids block

The volume of a $10 \times 6 \times 4$ cm cuboid:

Its surface area:

A cube of surface area 150 cm^2 has edge:

A cuboid of volume 120 cm^3 on a 5×4 base is high:

6 cm

24 cm

20 cm

4 cm

A cube of volume 512 cm^3 has edge:

7

8

9

171

A $60 \times 40 \times 50 \text{ cm}$ tank holds:

12

120

1200

120 000

A stone raising the level 1.5 cm on a 1000 cm^2 base has volume:

1500 cm^3

150 cm^3

666 cm^3

1.5 cm^3

A volume must be written in:

cm

cm²

cm³

any unit

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Volume	Hajm	Объём
Surface area	Sirt yuzasi	Площадь поверхности
Cuboid	Parallelepiped	Прямоугольный параллелепипед
Cube	Kub	Куб
Base area	Asos yuzasi	Площадь основания
Capacity	Sig‘im	Вместимость
Displacement	Siqib chiqarish	Вытеснение
Unit	Birlik	Единица измерения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The surface area of a cuboid is:

$$abc$$

$$2(ab + bc + ac)$$

$$6a^2$$

$$ab$$

The edge of a cube from its volume is:

$$V \div 3$$

the cube root of V

the square root of V

$$V \div 6$$

The edge of a cube from S is the square root of:

$$S$$

$$S \div 6$$

$$6S$$

$$S \div 2$$

1 m³ equals:

1000 cm³

10 000 cm³

1 000 000 cm³

100 cm³

A missing edge from the volume is found by dividing by:

three

the base area

the surface area

six

The most common lost mark in this block is:

arithmetic

the units

the diagram

the formula

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The volume of a $5 \times 4 \times 3$ cm cuboid

ANSWER 60 cm^3

2. Its surface area

ANSWER 94 cm^2

3. The volume of a cube of edge 4 cm

ANSWER 64 cm^3

4. Its surface area

ANSWER 96 cm^2

5. 1 litre in cm^3

ANSWER 1000 cm^3

6. The volume of an $8 \times 5 \times 2$ cm cuboid

ANSWER 80 cm^3

7. The number of faces of a cuboid

ANSWER 6

Medium · the standard question

7 PROBLEMS

1. The surface area of an $8 \times 5 \times 2$ cm cuboid

ANSWER 132 cm^2

2. The volume and surface area of a cube of edge 7 cm

ANSWER 343 cm^3 and 294 cm^2

3. $V = 120 \text{ cm}^3$ on a 5×4 base: the height

ANSWER 6 cm

4. A cube of volume 512 cm^3 : the edge

ANSWER 8 cm

5. A cube of surface area 150 cm^2 : the edge

ANSWER 5 cm

6. The capacity of a $60 \times 40 \times 50$ cm tank

ANSWER 120 litres

7. One face of a cube of surface area 294 cm^2

ANSWER 49 cm^2

Hard · stretch and reason

7 PROBLEMS

1. A cuboid 5 by 4 with $S = 94 \text{ cm}^2$: the third edge

ANSWER 3 cm

2. A tank with base $80 \times 50 \text{ cm}$ holding 120 litres: the depth

ANSWER 30 cm

3. A stone raising the level 1.5 cm on a 1000 cm^2 base

ANSWER 1500 cm^3

4. A 4 m by 3 m by 0.1 m slab at 2.4 t per m^3

ANSWER 2.88 tonnes

5. Card for an open box $8 \times 5 \times 4 \text{ cm}$

ANSWER 144 cm^2

6. Doubling every edge multiplies V and S by

ANSWER 8 and 4

7. Boxes $10 \times 8 \times 5 \text{ cm}$ in a crate $40 \times 32 \times 30 \text{ cm}$

ANSWER 96

07 Homework

Homework — 5 tasks

Write the unit on every line of working, not only on the answer.

- Find the volume and the surface area of a cuboid 9 cm by 5 cm by 4 cm.
- A cube has surface area 216 cm^2 . Find its edge and its volume.
- A cuboid has volume 252 cm^3 and a base 9 cm by 7 cm. Find its height.
- A tank 90 cm by 50 cm by 40 cm is filled at 12 litres a minute. How long does it take?
- List the five mistakes of the last section and write one line on how you will avoid each.